Assignment 4 — DBMS Fundamentals

Subject: **DBMS**
Assignment No.: **04**
Title: **Database Design for a Student Management System**
Difficulty: **Beginner**
Submission Format: **PDF / DOCX**

Objective

To understand fundamental DBMS concepts and design a simple database for managing student information.

Problem Statement

Educational applications need databases to store student information, notes, assignments, submissions, and grades efficiently.

Questions

1. What is a database?
2. What is DBMS?
3. Explain the advantages of DBMS.
4. Explain the major components of a DBMS.
5. What is a table?
6. Explain rows and columns.
7. What is a primary key?
8. What is a foreign key?
9. Explain different types of relationships.
10. Explain one-to-one, one-to-many, and many-to-many relationships.
11. What is data integrity?
12. Explain database constraints.
13. Explain database security.
14. Explain the role of DBMS in StudySphere.